Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

TECHNICAL CATALOGUE · MOTOR-CUM-HAND DRIVEN

CIE/777 HM

Mains motor with hand-crank operation

Sustained-voltage insulation testing

A universal, multi-purpose instrument for checking insulation and conductance under simulated operating conditions in high-tension electrical and electronic fields. It is ideally suited to the electrical maintenance of rotating plant machinery and distribution networks, communication cables and insulators, and to the servicing of control and switching equipment, transformers and industrial installations. The dielectric absorption test needs the test voltage applied for several minutes, and operating the instrument under motor power makes this easy. Polarisation index is the resistance after 10 minutes divided by the resistance after 1 minute, and better results are obtained with longer test periods. The high sensitivity of the instrument gives sensible readings in these tests. The CIE/777 can save several hours in locating intermittent shorts, defective electrical connections, and insulation and conductor failure, and in preventing costly repairs and breakdowns. The CIE/777 HM can also be operated by hand cranking, with all other details as for the CIE/777.



CIE/777 HM · ILLUSTRATIVE IMAGE

14	5	3 s
STANDARD RANGES	RATED VOLTAGES	RESPONSE TIME

STANDARD RATED TEST VOLTAGES

500 V · 1000 V · 2500 V · 5000 V · 10,000 V

01 HIGHLIGHTS

- ✓ Operated from 230 V, 50 Hz A.C. mains, or by hand cranking
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Anti-parallax mirror scale with knife-edge pointer (not provided on triple-range and above)
- ✓ Guard ring around the line terminal, plus a guard terminal
- ✓ ON-OFF switch, jewel lamp for mains ON and safety fuse
- ✓ Accurate and quick response: 3 seconds
- ✓ Collapsible generator handle
- ✓ Seasoned wooden cabinet with sunmica finish, detachable lid, carrying handle and lock
- ✓ Highly sensitive cross-coil indicator
- ✓ Externally accessible infinity adjuster
- ✓ Terminals accept standard cable terminations, including 4 mm plug and spade
- ✓ Conforms to IS 2992:1987

02 TYPICAL APPLICATIONS

- Field insulation testing where mains supply is uncertain or unavailable
- Sustained PI and dielectric absorption testing (motor-driven mode)
- Remote substation and railway traction equipment commissioning
- HV transformer and cable insulation assessment

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03 PRINCIPLE AND CONSTRUCTION

PRINCIPLE

A built-in motor operated from 230 V, 50 Hz A.C. mains, a static rectifier and a cross-coil ohm meter. The instrument can also be operated by hand cranking.

CONSTRUCTION

Housed in a seasoned wooden cabinet with sunmica finish, complete with detachable lid, carrying handle and lock. Terminals are designed to accommodate standard cable terminations, including 4 mm plug and spade. The weight is 12 kg including top cover (CIE/777: 9 kg).

04 SALIENT FEATURES

Mains motor drive

A built-in motor operated from 230 V, 50 Hz A.C. mains, with a static rectifier and a cross-coil ohm meter.

Guard ring and guard terminal

A guard ring around the line terminal and a guard terminal eliminate errors due to surface leakage. An externally accessible infinity adjuster is also provided.

Controls and protection

A highly insulated panel carries an ON-OFF switch, a jewel lamp indicating mains ON and a safety fuse protecting the motor, with a 3-pin socket for the supply cord.

Hand-crank operation

The instrument operates on A.C. mains as well as by hand cranking. A centrifugal clutch (governor) holds the output voltage constant, and the generator handle is collapsible.

05 PERFORMANCE AND TEST SPECIFICATION

Standard	IS 2992:1987
Supply	230 V, 50 Hz A.C. mains, or hand cranking
Indicator	Highly sensitive cross-coil indicator; anti-parallax mirror scale with knife-edge pointer (not provided on triple-range and above instruments)
Scale length	100 mm (approx.)
Accuracy	As per IS 2992:1987
Response time	3 seconds
High voltage test	Withstands minimum 2 kV or (rated voltage + 1) kV A.C. (r.m.s.) for 1 minute between the electrical circuit and the containing case
Insulation resistance of the instrument	Better than 50 MΩ at 500 V D.C. between the electrical circuit and the containing case

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06 CONSTRUCTION AND PHYSICAL SPECIFICATION

Hand operation	Collapsible generator handle; centrifugal clutch (governor) for constant output voltage
Guard and infinity adjuster	Guard ring around the line terminal and a guard terminal; externally accessible infinity adjuster
Panel	Highly insulated panel with ON-OFF switch, jewel lamp indicating mains ON and safety fuse protecting the motor; 3-pin socket for the supply cord
Supply cord	2 m, with 3-pin plug
Terminals	Designed to accommodate standard cable terminations, including 4 mm plug and spade
Body	Seasoned wooden cabinet with sunmica finish, detachable lid, carrying handle and lock
Weight (including top cover)	12 kg (approx.)
Dimensions (including top cover)	14 × 10 × 10 inch (approx.)

07 STANDARD RANGES BY RATED VOLTAGE

RATED VOLTAGE	INSULATION RESISTANCE RANGE (MΩ)
500 V	0–1000
1000 V	0–2000
2500 V	0–5000 · 0–10,000 · 0–20,000
5000 V	0–5000 · 0–10,000 · 0–20,000 · 50–50,000 · 100–100,000
10,000 V	0–10,000 · 0–20,000 · 50–50,000 · 100–100,000

Ranges shown are the standard builds. Any other range can be manufactured against specific requirements.

08 COMBINING VOLTAGES AND RANGES

Dual voltage and dual insulation range

Two rated voltages, each with its own insulation range. The selected range must be within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.

Single voltage, dual insulation range

One rated voltage with two insulation ranges, within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.

Dual voltage, single insulation range

Two rated voltages sharing one insulation range, within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.

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09 MULTIRANGE SERIES: CIE/777 HM/MR

SERIES	RATED VOLTAGE	INNER SCALE (MΩ)	OUTER SCALE (MΩ)
MR-A	5000 V	0–5000	50–100,000
	2500 V	0–2500	25–50,000
	1000 V	0–1000	10–20,000
MR-B	5000 V	0–2000	10–10,000
	2500 V	0–1000	5–5000
	1000 V	0–500	2–2000
MR-C	5000 V	100–100,000	Single scale
	2500 V	50–50,000	Single scale
MR-D	5000 V	0–5000	50–100,000
	2500 V	0–2500	25–50,000
MR-E	5000 V	50–50,000	Single scale
	2500 V	25–25,000	Single scale
	1000 V	10–10,000	Single scale

Multirange instruments are provided with a range selector switch, so one instrument covers every voltage and range listed in its series. Any other range can be manufactured against specific requirements.

10 SUPPLIED AND OPTIONAL

Supplied as standard

- ✓ Mains cord, 2 m
- ✓ 1 pair of H.V. testing leads

Optional, at extra cost

- Rexine carrying case
- 15-minute timer facility
- Testing leads, 1 m or 2 m
- Additional / special-grade H.V. testing leads, 2 m

11 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/777 HM, 500 V / 0–1000 MΩ**. Any range not listed can be manufactured against specific requirements.

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12 CIE HAND-DRIVEN AND MOTOR-OPERATED INSULATION TESTERS COMPARED

	CIE/444	CIE/666	CIE/555	CIE/777	CIE/777 HM
Drive	Hand-driven	Hand-driven	Hand-driven	Mains motor, 230 V, 50 Hz	**Mains motor, 230 V, 50 Hz, or hand cranking**
Body	Anti-magnetic cast aluminium	Seasoned wood, sunmica finish	Moulded bakelite	Seasoned wooden cabinet, sunmica finish	**Seasoned wooden cabinet, sunmica finish**
Rated voltages (V)	100, 250, 500, 1000, 2500, 5000	100, 250, 500, 1000, 2500, 5000, 10,000	100, 250, 500, 1000	500, 1000, 2500, 5000, 10,000	**500, 1000, 2500, 5000, 10,000**
Standard range combinations	27	32	11	14	**14**
Multirange series	MR-1 to MR-4	MR-1 to MR-8	Single range only	MR-A to MR-E	**MR-A to MR-E**
Scale length	90 mm	100 mm	75 mm	100 mm	**100 mm**
Response time	3 seconds	3 seconds	3 seconds	3 seconds	**3 seconds**
Weight (approx.)	2.5 kg	4.1 kg	1.4 kg	9 kg (incl. top cover)	**12 kg (incl. top cover)**
Conforms to	IS 2992:1987	IS 2992:1987	IS 2992:1987	IS 2992:1987	**IS 2992:1987**

Rated voltages, standard ranges and multirange series as listed in this catalogue series. Weights and scale lengths are approximate.

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